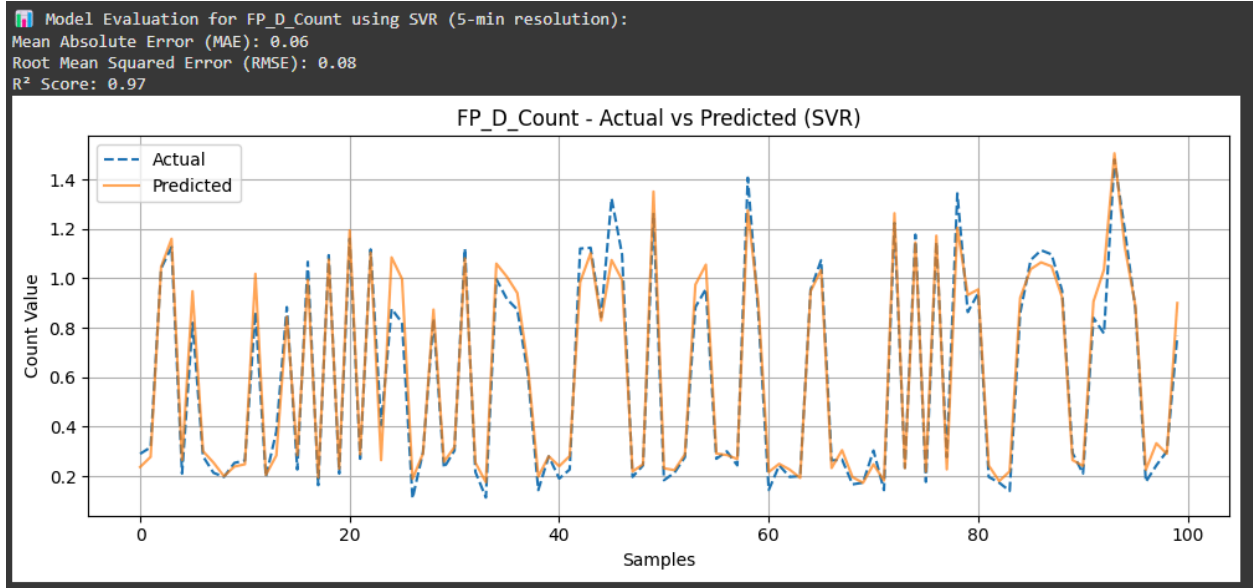
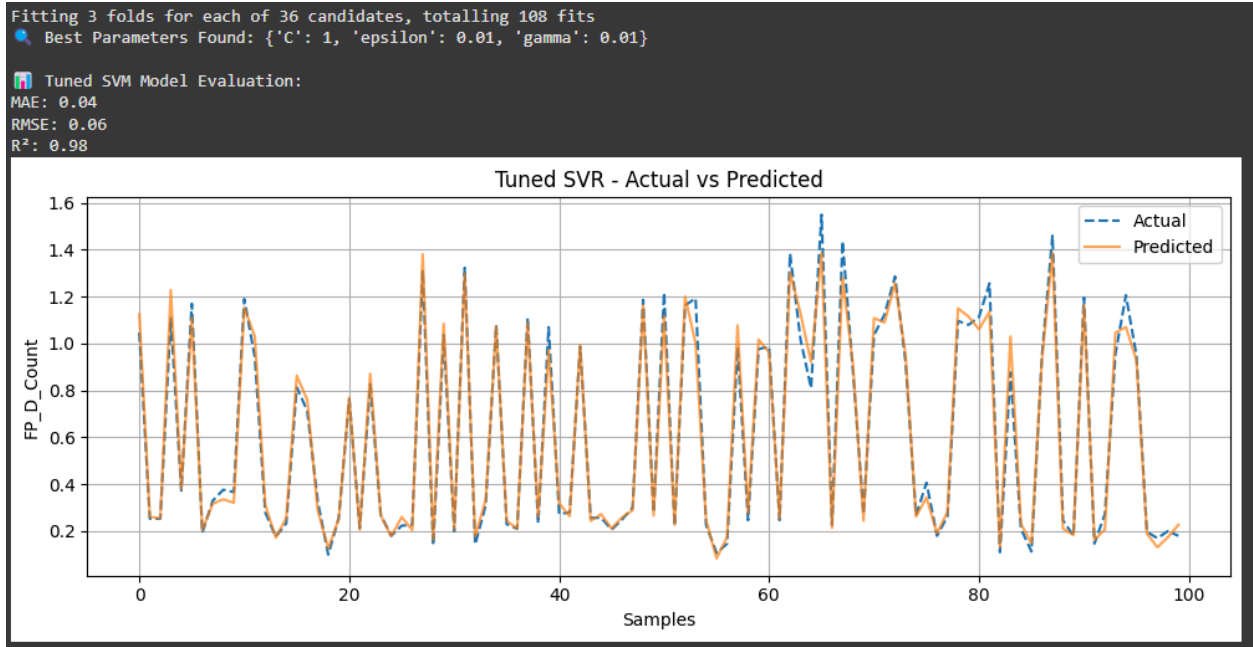


Regular, Lag = 2:



Using GridSearch:



Lag = 5, Print X_test, test size = 0.2:

Model Evaluation for FP_D_Count using SVR (5-min resolution, lag=5):

Mean Absolute Error (MAE): 0.06

Root Mean Squared Error (RMSE): 0.08

R² Score: 0.97

Features in X_test (first 5 rows):

	Time	FP_A_Count	FP_A_QueuePct	FP_A_Speed	FP_B_Count	FP_B_QueuePct	\
0	-0.853073	1.579128	1.157651	0.855123	0.301528	0.352240	
1	1.201290	2.058576	1.743238	0.768103	0.475170	0.525109	
2	-0.178326	0.989039	0.534048	1.373247	0.874547	0.987840	
3	0.365086	-1.160572	-1.227168	-1.080801	-1.070247	-1.120642	
4	-0.602452	-1.023587	-0.887772	-1.068179	-1.052882	-1.049049	

	FP_B_Speed	FP_C_Count	FP_C_QueuePct	FP_C_Speed	...	\
0	0.414416	0.709713	0.788653	1.187885	...	
1	1.240332	1.394497	1.440901	1.395850	...	
2	0.853028	0.761394	0.690248	0.836194	...	
3	-1.167041	-1.086230	-1.015439	-1.106907	...	
4	-1.095468	-0.737378	-0.763118	-0.563801	...	

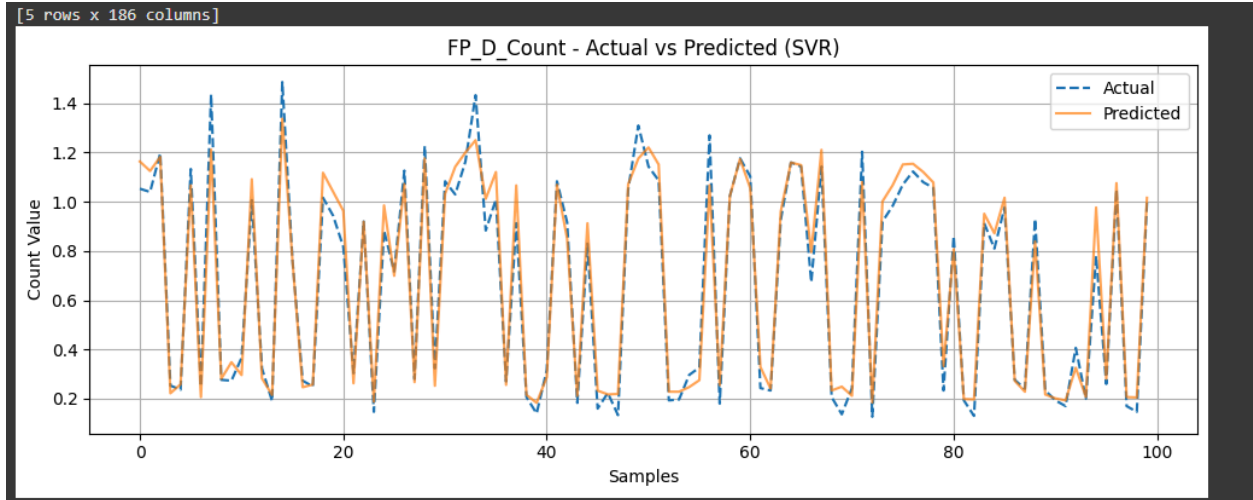
	Anand_Rao_Jnc_Count_lag5	Anand_Rao_Jnc_QueuePct_lag5	\
0	0.101670	0.120459	
1	0.197099	0.367627	
2	0.145806	-0.027112	
3	-0.389789	-0.344682	
4	-0.394560	-0.342426	

	Anand_Rao_Jnc_Speed_lag5	Race_Course_Jnc_Count_lag5	\
0	0.881184	2.165712	
1	0.873192	1.059743	
2	0.782055	0.373430	
3	-0.536691	-0.868053	
4	-0.481841	-0.557682	

	Race_Course_Jnc_QueuePct_lag5	Race_Course_Jnc_Speed_lag5	\
0	1.924135	1.274556	
1	1.200743	1.360966	
2	0.214340	0.014888	
3	-0.901130	-0.731217	
4	-0.640780	-0.623730	

	Palace_Road_Jnc_Count_lag5	Palace_Road_Jnc_QueuePct_lag5	\
0	-0.005828	-0.141923	
1	0.023183	-0.077106	
2	-0.161888	-0.210969	
3	-0.426989	-0.293518	
4	-0.395978	-0.285703	

	Palace_Road_Jnc_Speed_lag5	cosine_similarity
0	0.808327	-4.440892e-16
1	1.063874	6.661338e-16
2	0.993379	0.000000e+00
3	-0.589933	4.440892e-16
4	-0.012501	-2.220446e-16



Lag = 5, Print X_test, test size = 0.3:

Model Evaluation for FP_D_Count using SVR (5-min resolution, lag=5):

Mean Absolute Error (MAE): 0.06

Root Mean Squared Error (RMSE): 0.08

R² Score: 0.97

Features in X_test (first 5 rows):

	Time	FP_A_Count	FP_A_QueuePct	FP_A_Speed	FP_B_Count	FP_B_QueuePct	\
0	-0.853073	1.579128	1.157651	0.855123	0.301528	0.352240	
1	1.201290	2.058576	1.743238	0.768103	0.475170	0.525109	
2	-0.178326	0.989039	0.534048	1.373247	0.874547	0.987840	
3	0.365086	-1.160572	-1.227168	-1.080801	-1.070247	-1.120642	
4	-0.602452	-1.023587	-0.887772	-1.068179	-1.052882	-1.049049	

	FP_B_Speed	FP_C_Count	FP_C_QueuePct	FP_C_Speed	...	\
0	0.414416	0.709713	0.788653	1.187885	...	
1	1.240332	1.394497	1.440901	1.395850	...	
2	0.853028	0.761394	0.690248	0.836194	...	
3	-1.167041	-1.086230	-1.015439	-1.106907	...	
4	-1.095468	-0.737378	-0.763118	-0.563801	...	

	Anand_Rao_Jnc_Count_lag5	Anand_Rao_Jnc_QueuePct_lag5	\
0	0.101670	0.120459	
1	0.197099	0.367627	
2	0.145806	-0.027112	
3	-0.389789	-0.344682	
4	-0.394560	-0.342426	

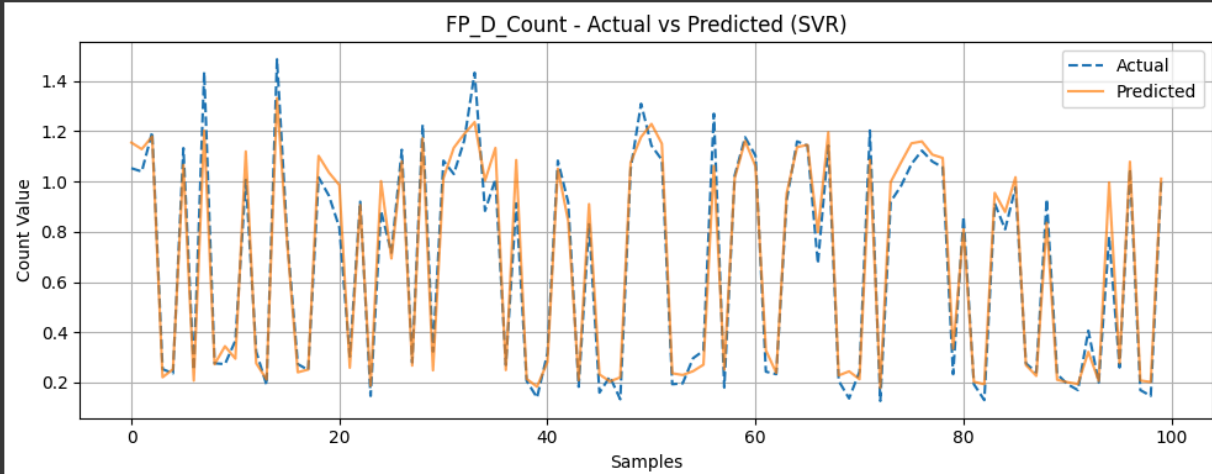
	Anand_Rao_Jnc_Speed_lag5	Race_Course_Jnc_Count_lag5	\
0	0.881184	2.165712	
1	0.873192	1.059743	
2	0.782055	0.373430	
3	-0.536691	-0.868053	
4	-0.481841	-0.557682	

	Race_Course_Jnc_QueuePct_lag5	Race_Course_Jnc_Speed_lag5	\
0	1.924135	1.274556	
1	1.200743	1.360966	
2	0.214340	0.014888	
3	-0.901130	-0.731217	
4	-0.640780	-0.623730	

	Palace_Road_Jnc_Count_lag5	Palace_Road_Jnc_QueuePct_lag5	\
0	-0.005828	-0.141923	
1	0.023183	-0.077106	
2	-0.161888	-0.210969	
3	-0.426989	-0.293518	
4	-0.395978	-0.285703	

	Palace_Road_Jnc_Speed_lag5	cosine_similarity
0	0.808327	-4.440892e-16
1	1.063874	6.661338e-16
2	0.993379	0.000000e+00
3	-0.589933	4.440892e-16
4	-0.012501	-2.220446e-16

[5 rows x 186 columns]



10-minute interval:

10-Min Interval Model Evaluation:

MAE: 0.06, RMSE: 0.08, R²: 0.96

X_test sample:

	Time	FP_A_Count	FP_A_QueuePct	FP_A_Speed	FP_B_Count	FP_B_QueuePct	\
0	0.084490	-0.688971	-0.761325	-0.558963	-0.863820	-0.911817	
1	0.337961	-1.101366	-1.014556	-1.074795	-0.882727	-0.870177	
2	-0.212433	1.402848	1.724025	1.076187	1.906036	1.987364	
3	1.675322	-0.710676	-0.752197	-0.595317	-0.599124	-0.613239	
4	0.620400	-1.060669	-1.099335	-1.017757	-0.873273	-0.885319	

	FP_B_Speed	FP_C_Count	FP_C_QueuePct	FP_C_Speed	...	\
0	-0.871726	-0.720266	-0.544292	-0.737575	...	
1	-0.862972	-1.007417	-0.929218	-1.073009	...	
2	1.756149	1.392349	2.754735	1.510933	...	
3	-0.491055	-0.563016	-0.496049	-0.684420	...	
4	-0.966159	-1.096298	-1.001412	-1.272569	...	

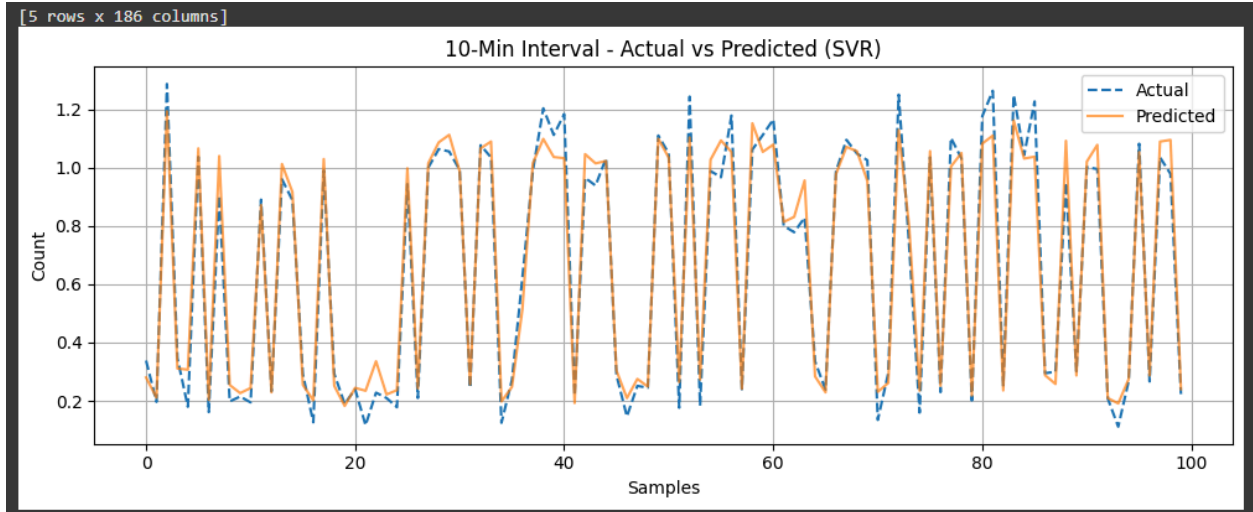
	Anand_Rao_Jnc_Count_lag5	Anand_Rao_Jnc_QueuePct_lag5	\
0	-0.519312	-0.445386	
1	-0.464012	-0.439206	
2	0.098863	0.060406	
3	-0.483104	-0.436898	
4	0.133097	0.117436	

	Anand_Rao_Jnc_Speed_lag5	Race_Course_Jnc_Count_lag5	\
0	-1.212914	-1.163170	
1	-0.860077	-0.788531	
2	1.221142	0.458716	
3	-0.990212	-0.679164	
4	1.271601	0.547141	

	Race_Course_Jnc_QueuePct_lag5	Race_Course_Jnc_Speed_lag5	\
0	-1.213755	-1.597562	
1	-0.815864	-0.763048	
2	0.509346	0.915047	
3	-0.716158	-0.591988	
4	0.609755	0.982985	

	Palace_Road_Jnc_Count_lag5	Palace_Road_Jnc_QueuePct_lag5	\
0	-0.422321	-0.306923	
1	-0.432338	-0.290668	
2	-0.092283	-0.213582	
3	-0.409222	-0.303792	
4	-0.168307	-0.229327	

	Palace_Road_Jnc_Speed_lag5	cosine_similarity
0	-1.012448	2.220446e-16
1	-0.698457	-1.110223e-16
2	0.934864	-1.110223e-16
3	-0.169408	2.220446e-16
4	1.079594	4.440892e-16



30-minute interval:

30-Min Interval Model Evaluation:
MAE: 0.07, RMSE: 0.08, R²: 0.96

X_test sample:

	Time	FP_A_Count	FP_A_QueuePct	FP_A_Speed	FP_B_Count	FP_B_QueuePct	\
0	1.006415	-0.917702	-0.863668	-0.828818	-0.901958	-0.885592	
1	1.349179	-0.880827	-0.769845	-0.914204	-1.098860	-1.109385	
2	-1.662772	0.141530	0.121992	0.369966	0.249422	0.244554	
3	-1.196029	1.120559	1.197935	1.042882	0.469685	0.354031	
4	0.853265	1.007168	1.301082	1.080104	0.980297	0.974343	

	FP_B_Speed	FP_C_Count	FP_C_QueuePct	FP_C_Speed	...	\
0	-0.778385	-0.904983	-0.844323	-0.867761	...	
1	-1.097461	-1.095507	-1.058031	-1.084117	...	
2	0.316744	0.173858	0.195438	0.191363	...	
3	0.525402	0.754956	0.596869	1.046079	...	
4	1.068251	1.707575	1.594740	1.161568	...	

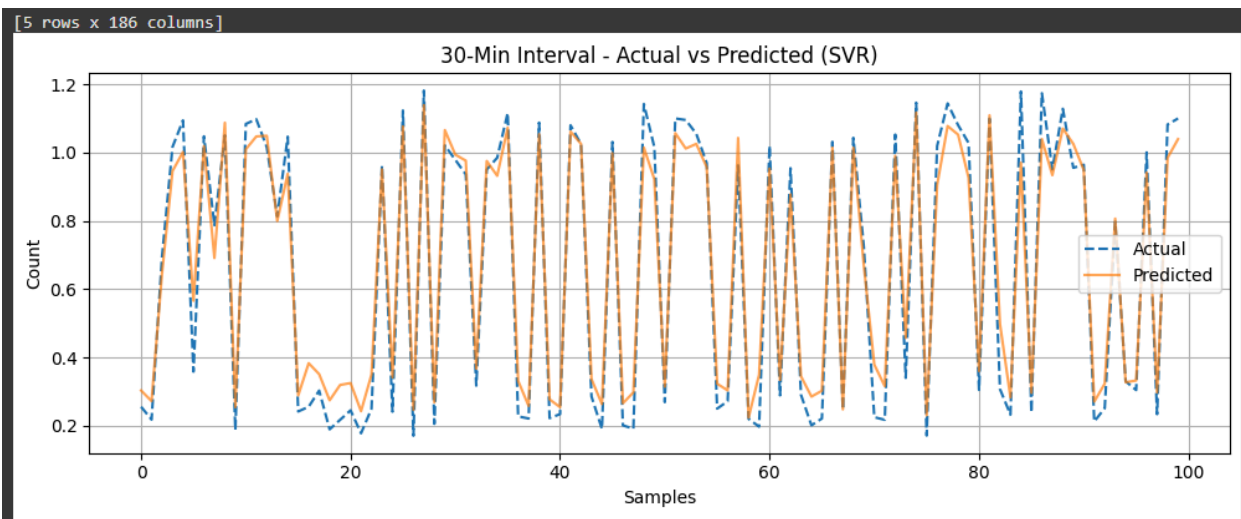
	Anand_Rao_Jnc_Count_lag5	Anand_Rao_Jnc_QueuePct_lag5	\
0	-0.619203	-0.556320	
1	-0.520117	-0.432954	
2	-0.649973	-0.598845	
3	1.160388	0.806587	
4	0.197760	0.084508	

	Anand_Rao_Jnc_Speed_lag5	Race_Course_Jnc_Count_lag5	\
0	-0.910216	-1.031673	
1	-0.691463	-0.683079	
2	-1.191725	-0.996647	
3	0.461563	0.678776	
4	1.207765	1.260879	

	Race_Course_Jnc_QueuePct_lag5	Race_Course_Jnc_Speed_lag5	\
0	-1.048135	-1.095991	
1	-0.631444	-0.733377	
2	-1.045322	-1.080929	
3	0.971859	0.911139	
4	1.050382	1.181874	

	Palace_Road_Jnc_Count_lag5	Palace_Road_Jnc_QueuePct_lag5	\
0	-0.463911	-0.362229	
1	-0.413596	-0.351414	
2	-0.511500	-0.379461	
3	-0.080382	-0.224910	
4	-0.082263	-0.228587	

	Palace_Road_Jnc_Speed_lag5	cosine_similarity
0	-0.569230	2.220446e-16
1	-0.218825	-5.551115e-16
2	-1.285205	0.000000e+00
3	1.274557	-2.220446e-16
4	1.177180	8.881784e-16



45-minute interval:

45-Min Interval Model Evaluation:

MAE: 0.07, RMSE: 0.09, R²: 0.95

X_test sample:

	Time	FP_A_Count	FP_A_QueuePct	FP_A_Speed	FP_B_Count	FP_B_QueuePct	\
0	0.538863	2.042156	1.841167	0.972838	0.890359	0.937577	
1	-1.363653	-1.052243	-0.975313	-1.141065	-1.293717	-1.314958	
2	0.065983	-0.858999	-0.850660	-0.722233	-0.932732	-0.956920	
3	-0.868779	1.007771	0.754575	1.120079	0.972092	0.943258	
4	-0.703821	-1.112905	-1.226378	-1.235236	-1.089386	-1.062025	

	FP_B_Speed	FP_C_Count	FP_C_QueuePct	FP_C_Speed	...	\
0	1.013096	1.374192	1.321735	0.963134	...	
1	-1.346261	-1.220269	-1.146785	-1.345688	...	
2	-0.920838	-0.875742	-0.799305	-0.858148	...	
3	1.043946	1.162301	1.275118	1.146065	...	
4	-1.099068	-1.008377	-0.958673	-1.003827	...	

	Anand_Rao_Jnc_Count_lag5	Anand_Rao_Jnc_QueuePct_lag5	\
0	0.263760	0.136877	
1	-0.666136	-0.592222	
2	-0.709842	-0.648038	
3	1.721665	1.024166	
4	-0.676334	-0.610220	

	Anand_Rao_Jnc_Speed_lag5	Race_Course_Jnc_Count_lag5	\
0	1.297588	1.025490	
1	-0.776157	-0.962473	
2	-1.039285	-1.023650	
3	0.362475	0.907139	
4	-0.839412	-0.911016	

	Race_Course_Jnc_QueuePct_lag5	Race_Course_Jnc_Speed_lag5	\
0	0.898242	1.220515	
1	-0.909041	-0.915639	
2	-1.021943	-1.183990	
3	0.850695	0.788105	
4	-0.965973	-0.838022	

	Palace_Road_Jnc_Count_lag5	Palace_Road_Jnc_QueuePct_lag5	\
0	-0.042954	-0.204505	
1	-0.490686	-0.408438	
2	-0.531133	-0.418183	
3	-0.075192	-0.251672	
4	-0.499550	-0.403217	

	Palace_Road_Jnc_Speed_lag5	cosine_similarity
0	1.151402	-4.440892e-16
1	-0.371618	-2.220446e-16
2	-1.258970	-7.771561e-16
3	1.321687	0.000000e+00
4	-0.794622	2.220446e-16

[5 rows x 186 columns]

